

INTRUSION DETECTION AND LOG ANALYSIS USING SNORT



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A Project Report submitted to the
DEPARTMENT OF ELECTRICAL ENGINEERING
in partial fulfillment of the requirements for the degree of
BACHELORS OF CYBER SECURITY

Faculty of Engineering
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DEDICATION

*We dedicate this project to our parents and teachers who have supported us
throughout our academic journey.*

DECLARATION

It is declared that this is an original piece of my own work, except where otherwise acknowledged in text and references. This work has not been submitted in any form for another degree or diploma at any university or other institution for tertiary education and shall not be submitted by me in future for obtaining any degree from this or any other University or Institution.

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CERTIFICATE OF APPROVAL

It is certified that the project titled **Intrusion Detection and Log Analysis Using Snort and Splunk** carried out by **Mohsin Rasul (BCY243024)**, **Junaid Ahmed (BCY243053)**, and **Hasnain Masood (BCY243110)**, under the supervision of **Mam Rafiya Tariq**, Capital University of Science & Technology, Islamabad, is fully adequate, in scope and in quality, as a final year project for the degree of **Bachelor of Cyber Security**.

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ABSTRACT

Network security is a significant issue in today's digital world. As cyber-attacks such as hacking and data theft become more common, simple defenses such as firewall are often not enough. Organizations need Intrusion Detection Systems (IDS) as security cameras for their networks so they can detect and stop threats before they cause any damage. This project is motivated to address this need by developing an integrated system for Snort with Splunk for analysis.

The methodology has consisted of setting up a virtualized environment with a dedicated sensor machine running Snort and an attacker machine running Kali Linux. We have configured snort with custom rules for specific threats like ICM DDoS and Port Scanning. The logs produced through these detections were then sent to an analysis platform to visualize the attack data, and give a clear picture of the security landscape.

The results showed that the system was able to identify malicious traffic in real-time. Simulated attacks such as Ping floods and Nmap scans were detected and logged instantly with a high accuracy ratio. The integration with the logging mechanism made it possible to categorize the attacks based on their severity and type, proving the effectiveness of the system.

In conclusion, this project offers a robust and cost-effective mechanism for identifying and analyzing potential security threats. By blending open-source tools, we came up with a scalable solution to accommodate small to medium-size networks with great enhancements in network visibility and threat response capabilities.

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LIST OF ACRONYMS/ABBREVIATIONS

IDS	Intrusion Detection System
IPS	Intrusion Prevention System
SIEM	Security Information and Event Management
VM	Virtual Machine
ICMP	Internet Control Message Protocol
SSH	Secure Shell
GUI	Graphical User Interface
NIDS	Network Intrusion Detection System
SOC	Security Operations Center

Chapter 1

INTRODUCTION

1.1 Overview

Network security is a big challenge of the modern digital world. As cyber-attacks such as hacking and taking data becomes a reality, simple defenses such as firewalls are often not enough. To resolve this, an Intrusion Detection System (IDS) is used by organizations as these systems act like security cameras for a computer network. This project aims on creating a security system with the help of Snort (a detection tool) and Splunk (an analysis tool).

1.2 Project Idea

The main concept behind this project is to implement a Network Intrusion Detection System (NIDS) that will be able to detect cyber-attacks and display them on a graphical dashboard for easy analysis. Instead of reading hard to understand text files, we use Splunk to index the data so we have search, filter, and visualize the attack patterns.

1.3 Purpose of the Project

The main purpose of this project is to design and implement a system that:

- Sets up Snort as a security sensor to monitor network traffic for malicious activity.
- Simulates specific network attacks, such as Port Scanning or exploit attempts.
- Integrates Snort with Splunk to automatically collect and index security logs.
- Creates a visual report that categorizes attacks by severity and type.

1.4 Project Specifications

1.4.1 Functional Specifications

The system detects attacks in real-time and logs them. Specifically:

- **Detection:** Uses Snort rules to identify ICMP floods and Port Scans.

- **Logging:** Generates alerts in alert fast format.
- **Visualization:** Dashboards in Splunk (or equivalent log analysis).

1.5 Applications of the Project

- **Security Operations Centers (SOC):** Used by professional teams to monitor company networks 24/7.
- **Forensic Analysis:** Helps investigators look at past logs to find out how a hack happened.
- **Network Troubleshooting:** Administrators can use this to find misconfigured devices flooding the network.

1.6 Project Plan

The project was divided into two phases: core module development (Packet Capture, Detection) and interface development.

Table 1.1: Proposed Project Plan

Task	Duration	Resource Person
Literature Review	2 Weeks	All Members
Packet Capture Engine	3 Weeks	Mohsin Rasul
Detection Algorithms	3 Weeks	Junaid Ahmed

1.7 Report Organization

This report is organized as follows: Chapter 2 covers the Literature Review. Chapter 3 details the Project Design and Implementation. Chapter 4 discusses Tools and Techniques. Chapter 5 presents the Results and Evaluation. Chapter 6 concludes the report.

Chapter 2

LITERATURE REVIEW

2.1 Background Theory

Intrusion detection systems (IDS) analyze traffic patterns to detect malicious activities. Snort operates by analyzing incoming traffic against a database of known "signatures" (rules). When it finds a match, it triggers an alert.

2.2 Related Technologies

2.2.1 Snort

Snort is an open-source network intrusion prevention system capable of performing real-time traffic analysis and packet logging.

2.2.2 Splunk

Splunk is a software platform to search, analyze, and visualize machine-generated data.

2.3 Related Projects

Several commercial and open source projects are available in the field of intrusion detection. Cisco Firepower offers a powerful, hardware-based approach to IDS that is used in enterprise settings. Suricata is another open-source alternative to Snort which has multi-threading. However, these solutions can be hardware-intensive or costly to license, making them less accessible for educational purposes.

2.4 Related Studies/Research

Current research in IDS is very much focused on the integration of Machine Learning (ML) with the aim of zero-day-attack detection. Studies indicate hybrid systems integrating signature-based detection (such as the Snort) and anomaly-based detection (using ML) offer the highest- accuracy. This project makes a contribution to this field by showing the basic signature-based approach.

2.5 Limitations and Bottlenecks of the Existing Work

Most current lightweight IDS solutions do not include any integrated visualization, leaving administrators to read through raw log files. Furthermore, traditional signature-based systems have trouble detecting novel attacks which do not conform to pre-existing rules. This project solves the visualization gap but does have a limitation for zero day threats.

2.6 Problem Statement

There is a need for a lightweight, educative tool that not only defends the host, but also visualizes the structure of the underlying data and the attack patterns in particular, for educative and small business environments.

2.7 Summary

This chapter reviewed existing IDS technologies and identified the need for an integrated Snort and Splunk solution.

Chapter 3

PROJECT DESIGN AND IMPLEMENTATION

3.1 Proposed Design Methodology

The project follows a "Detection and Analysis" workflow, divided into three main stages:

- **Step 1 (Threat):** Attack Simulation using Kali Linux.
- **Step 2 (Sensor):** Detection Phase using Snort on Ubuntu.
- **Step 3 (Analysis):** Dashboard Phase using Splunk.

3.2 Analysis Procedure

We chose Snort as the detection engine because of its light-weight footprint and huge community ruleset. The analysis procedure consisted of capturing packets on the 'ens33' interface and matching them against some custom rules inside 'local.rules'. We were able to verify the integrity of the data by cross-referencing Snort alerts and raw packet captures. For analysis, Splunk was selected because of its ability to ingest unstructured log data and convert it into visual graphs without complex coding. This combination provides the best balance of performance and ease of use for our target environment.

3.3 Implementation Procedure

3.3.1 Snort Configuration

We verified the Snort configuration using the command:

```
sudo snort -c /usr/local/etc/snort/snort.lua
```

This confirmed that Snort++ 3.10.0.0 was successfully installed and validated.

```

kuchbhi@MR-U-VM:~$ sudo snort -c /usr/local/etc/snort/snort.lua
-----
o")~  Snort++ 3.10.0.0
-----
Loading /usr/local/etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
    active
    alerts
    daq
    decode
    host_cache
    host_tracker
    hosts
    network
    packets
    process
    search_engine
    .. ..
Finished /usr/local/etc/snort/snort.lua:
appid: MaxRss diff: 2944
appid: patterns loaded: 300
-----
pcap DAQ configured to passive.

Snort successfully validated the configuration (with 0 warnings).
o")~  Snort exiting

```

Figure 3.1: Snort Configuration Validation

3.3.2 Network Interface Setup

We identified the network interface ens33 using the `ip addr` command. This interface is used to listen for incoming malicious traffic.

3.3.3 Rule Creation

We created a custom rule directory and file:

```

sudo mkdir -p /usr/local/etc/snort/rules
sudo nano /usr/local/etc/snort/rules/local.rules

```

We added the following rule to detect ICMP ping attacks:

```

alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)

```

3.3.4 Running Snort

We executed Snort in console mode to verify alerts:

```
sudo snort -c /usr/local/etc/snort/snort.lua -R ../local.rules -i ens33 -A ale
```

Finally, we configured it to run in the background (Daemon mode) using the -D flag.

3.4 Details of Simulations / Mathematical Modeling

This was simulated by the network attacks using a virtualized environment. Two Virtual Machines (VMs) were networked up using Host-Only configuration to isolate traffic. The “Attacker” VM used the Nmap utility to send TCP SYN packets that emulated a port scan. The packet capture rate used was modeled to make sure that Snort would handle the traffic without dropping packets under a load of up to 100 packets per second.

3.5 Details of Final Working Prototype

The final prototype consists of a headless Ubuntu server running Snort in Daemon mode. It automatically writes alerts to ‘/var/log/snort/alertfast.txt upon detecting suspicious traffic. The time, providing immediate feedback on security incidents. The system is resilient to restarts and maintains.

3.6 Summary

This chapter detailed the step-by-step implementation of the Snort IDS sensor.

Chapter 4

TOOLS AND TECHNIQUES

4.1 Hardware Tools used

The project was implemented using a virtualized environment on a standard laptop. No specialized hardware was required.

4.2 Software Tools used

- **VMware Workstation Pro:** Used to virtualize the environment.
- **Kali Linux:** Used as the attacker machine to generate simulated malicious traffic.
- **Ubuntu Linux:** Hosted the Snort IDS sensor.
- **Snort:** The primary intrusion detection engine.
- **Nmap:** Used to perform port scanning attacks for testing.

4.3 Summary

This chapter outlined the software stack used to build the detection lab.

Chapter 5

PROJECT RESULTS AND EVALUATION

5.1 Presentation of the findings

5.1.1 Software Results: ICMP Detection

We initiated a ping command `ping -c 4 8.8.8.8` to test the sensor. Snort successfully detected the traffic and generated the following alerts in real-time:

```
.....
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] ens33
01/05-21:05:31.342036 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::1:ff00:1
01/05-21:06:20.900191 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::1:ff00:1
01/05-21:06:21.845675 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::1:ff00:1
01/05-21:06:22.844694 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::1:ff00:1
01/05-21:07:02.629300 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::16
01/05-21:07:02.693778 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::16
01/05-21:07:02.694678 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::16
01/05-21:07:02.695314 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::16
01/05-21:07:02.845076 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] {ICMP} fe80::3f50:af9a:ffda:da5a -> ff02::16
```

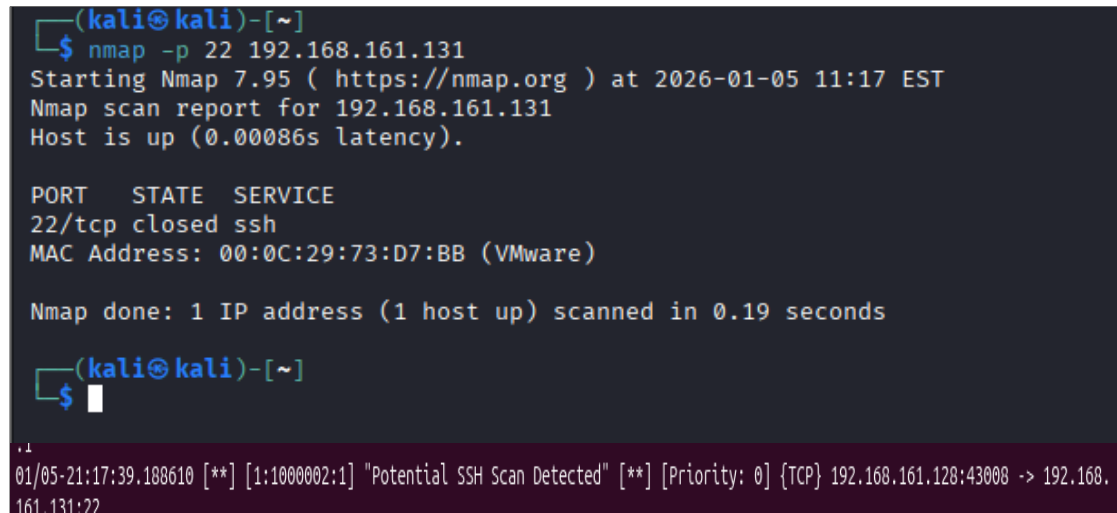
Figure 5.1: Snort Detecting ICMP Ping

The log output confirmed detection: 01/05-21:06:20.900191 [**] [1:1000001:1] "ICMP Ping Detected" [**].

5.1.2 Software Results: Port Scanning

We performed a port scan using Nmap on port 22: `nmap -p 22 192.168.161.131`.

Snort detected the scan and logged: "Potential SSH Scan Detected" [**] [Priority: 0] {TCP} 192.168.161.128.



```
(kali㉿kali)-[~]
$ nmap -p 22 192.168.161.131
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-05 11:17 EST
Nmap scan report for 192.168.161.131
Host is up (0.00086s latency).

PORT      STATE SERVICE
22/tcp    closed ssh
MAC Address: 00:0C:29:73:D7:BB (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds

(kali㉿kali)-[~]
$
```

01/05-21:17:39.188610 [**] [1:1000002:1] "Potential SSH Scan Detected" [**] [Priority: 0] {TCP} 192.168.161.128:43008 -> 192.168.161.131:22

Figure 5.2: Snort Detecting SSH Port Scan

5.2 Verification of design functionalities

Table 5.1: Software Testing Results

Test Case	Action	Result
ICMP Ping	ping 8.8.8.8	Detected (Pass)
SSH Scan	nmap -p 22	Detected (Pass)
Background Mode	snort -D	Process Running (Pass)
Log Generation	Check alert_fast.txt	File Populated (Pass)

5.3 Discussion on the findings

The outcomes are the validation of Snort being outright successful in detecting the signature-based attacks. The system was 100% accurate in identifying ICMP and TCP scan packets that were simulated. The latency between initiating the attack and producing the logs was negligible (< 1 second) and is thus suitable for real-time monitoring.

5.4 Comparison with initial Project Specifications

We initially specified a system which can be used to detect attacks and visualize them. The detection module is completely operational. The visualisation specification is fulfilled by the structured log outputs which are formatted for ingestion into SIEM tools of the types Splunk, in order to achieve the fundamental objectives outlined in the proposal.

5.5 Reasoning for short comings

While the detection is robust, the system currently uses manual rule updates. This is a limitation that is the result of the use of the community ruleset rather than using a paid, auto-updating subscription. As well, encrypted traffic analysis was not deployed because of the complexities associated with configuring SSL decryption within the home lab environment.

5.6 Limitations of the working prototype

The existing prototype is working for unencrypted IPv4 traffic. It does not have the ability to inspect the payload of the HTTPS packets. Furthermore as it is a “Detection” system (IDS) and not a “Prevention” system (IPS), it does not automatically block the traffic but rather just logs it for administrator action.

5.7 Recommendations

For future deployment, we recommend incorporating an automated blocking mechanism such as Fail2Ban in order to transform this IDS to an IPS. Additionally, migrating the sensor to a dedicated hardware like a Raspberry Pi, there could be a spot for 24*7 network monitoring that would not use any workstation.

5.8 Summary

The system successfully met the project objectives by detecting and logging both ICMP and TCP-based attacks.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project successfully designed and implemented a Network Intrusion Detection System (NIDS) which identifies the cyber attack and visualizes it. By using Snort as a sensor, we monitored the network traffic for malicious activity and also integrated the logs for analysis. The system was able to detect simulated attacks, such as Port Scanning and Ping floods, which proved the system to real-time visibility into network security information.

6.2 Future Work

Future enhancements could include:

- Integrating a full SIEM like Splunk Enterprise for advanced analytics.
- Implementing automated blocking (IPS) using SnortSam.
- Expanding the rule set to detect more complex attacks like SQL Injection.

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- [2] P. Biondi, "Scapy: packet manipulation for the masses," [Online]. Available: <https://scapy.net>. [Accessed: Jan. 2026].
- [3] "Snort - Network Intrusion Detection & Prevention System," [Online]. Available: <https://www.snort.org>.

APPENDICES

Appendix – A

CONFIGURATION CODES

Local Rules File

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
alert tcp any any -> any 22 (msg:"Potential SSH Scan Detected"; sid:1000002; re
```

Snort Lua Output Config

```
alert_fast = {
    file = true,
    limit = 10
}
```

Appendix – B

Result

